

JEE(Main + Advance) : Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 13-05-2021**ANSWER KEY****PAPER-2****PHYSICS**

SEC-1	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	A,B,C	A,B,C,D	A,B	A,B,C	B,D	A,B	A,B
SEC-2	Q.	1	2	3	4				
	A.	D	A	B	A,B,C				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	3	1	5	6	3	2	8	4

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B	A,B,C	A,B,D	A,C,D	B,D	A,B,D	A,C
SEC-2	Q.	1	2	3	4				
	A.	C	A	A,C	A,B,D				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	9	3	2	4	1	6	3	6

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A,C	A,C,D	A,B,C	A,B	B,D	A,B	B,C,D
SEC-2	Q.	1	2	3	4				
	A.	B	C	A,B,C	A				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	7	2	2	1	8	6	4	1

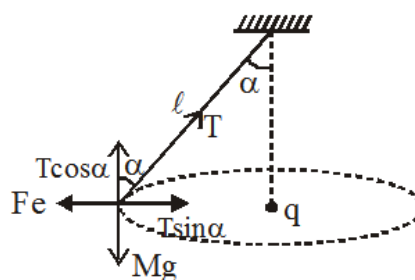
HINT – SHEET**PHYSICS****SEC-1**1. **Ans (A,B,C,D)**

- (A) $\vec{F}_{\text{net}} = \vec{0} \Rightarrow$ particle is in equilibrium
 (B) Bottom most point is AOR, so no point has vertical velocity
 (C) Because of angular momentum conservation, as 'I' increases 'w' must decrease.
 (D) The resultant of centripetal force and tangential force will be skew.

2. **Ans (A,B,C)**

$$T \cos \alpha = Mg$$

$$T \sin \alpha - F_c = \frac{Mv^2}{r}$$



3. **Ans (A,B,C,D)**

In standing wave energy is confined between node & antinode.

Node and antinode do not transfer its energy but element between node & antinode transfers its energy to other element.

4. **Ans (A,B)**

$$(A) \lambda_i = \frac{10u - \left(\frac{u}{2}\right)}{f} = \frac{19u}{2f}$$

$$(B) f_i = \frac{10u - u}{10u - \frac{u}{2}} \cdot f = \frac{18f}{19}$$

$$(C) \lambda_r = \frac{10u + u}{f_r} = \frac{11u \times 19}{18f} = \frac{11 \times 19u}{18f}$$

$$(D) f_r = \frac{18f}{19}$$

5. **Ans (A,B,C)**

$TV^{\gamma-1} = \text{constant}$

$$\Rightarrow V^{\gamma-1} + T(\gamma-1)V^{\gamma-2} \frac{dV}{dt} = 0$$

$$\Rightarrow \frac{dV}{dT} = -\frac{V}{T(\gamma-1)}$$

$$\Rightarrow \left| \frac{dV}{dT} \right|_{T_0, V_0} = \frac{V_0}{T_0(\gamma-1)} = m \Rightarrow$$

$$\gamma - 1 = \frac{V_0}{T_0 m}$$

$$\Rightarrow C_v = \frac{R}{\gamma-1} = \frac{Rt_0 m}{V_0} \Rightarrow C_p = C_v + 1 =$$

$$\left(1 + \frac{T_0 m}{V_0}\right) R$$

6. **Ans (B,D)**

Use Ampere's circuital law.

7. **Ans (A,B)**

$$\frac{R}{R_1 + R_2} = \frac{L/4}{3L/4}$$

$$3R = (R_1 + R_2)$$

$$\frac{R + R_1}{R_2} = \frac{2/3}{1/3}$$

$$3R = R_2 + R_1$$

$$R = 2R_2 - R_1$$

$$4R = 3R_2$$

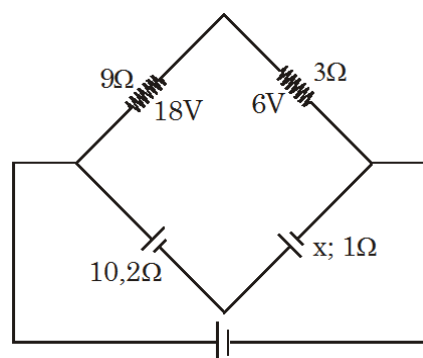
$$R_1 = \frac{5R}{3}$$

$$R_2 = \frac{4R}{3}$$

PHYSICS

SEC-2

1. **Ans (D)**



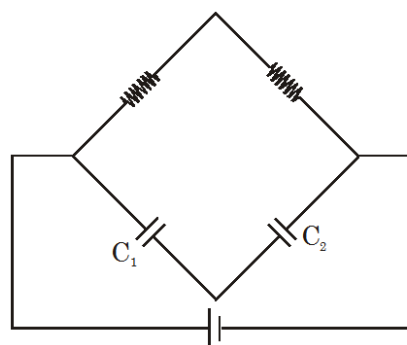
$$18 = 10 + 2i$$

$$i = 4A$$

$$x + 4 = 6$$

$$x = 2 \text{ volt}$$

2. **Ans (A)**



$$V_1 = \frac{q}{C_1} = 18$$

$$V_2 = \frac{q}{C_2} = 6$$

$$\Rightarrow \frac{C_1}{C_2} = \frac{1}{3}$$

4. **Ans (A,B,C)**

Particle would travel in cycloidal path.

$$x = vt - R \cos \omega t$$

$$y = R(1 - \cos \omega t)$$

$$\text{where, } v = \frac{E_0}{B_0}, \omega = \frac{qB_0}{m}, R = \frac{mE_0}{qB_0^2}$$

$$y_{\max} = 2R = \frac{2mE_0}{qB_0^2}$$

and radius of curvature at $y_{\max}$

$$= 4R = \frac{4mE_0}{qB_0^2}$$

PHYSICS

SEC-3

1. **Ans (3)**

$T \propto \sqrt{\text{length}}$ free length becomes $\frac{\ell}{4}$ So time

period becomes $T_1 = \frac{T}{2}$

Half oscillation is with free length ℓ and half with free length $\frac{\ell}{4}$

$$\text{Total time} = \frac{T}{2} + \frac{T_1}{2} = \frac{3T}{4}$$

2. **Ans (1)**

$$\Delta x = (\mu_{\text{rel}} - 1)t = \left[\frac{\mu_2}{\mu_1} - 1 \right] t$$

$$\mu_1 = \frac{4}{3}$$

$$\mu_2 = \frac{3}{2}$$

$$t = 8 \mu\text{m}$$

$$\Delta x = 1 \mu\text{m}$$

3. **Ans (5)**

Temp. is increased by $\Delta\theta$ then

$$\Delta\ell = \ell\alpha\Delta\theta$$

$$\Rightarrow \Delta\theta = \frac{\Delta\ell}{\ell\alpha}$$

$$E_1 = (\rho A \ell) S \Delta\theta = \rho A \ell S \frac{\Delta\ell}{\ell\alpha}$$

$$\text{when stretched, Stress} = Y \frac{\Delta\ell}{\ell}$$

$$E_2 = \frac{1}{2} \left(Y \frac{\Delta\ell}{\ell} \right) \left(\frac{\Delta\ell}{\ell} \right) \times A \ell = \frac{Y (\Delta\ell)^2 A}{2\ell}$$

$$\text{So, } \frac{E_1}{E_2} = \frac{E_1}{E_2} = \frac{\rho A \ell S \Delta\ell \times 2\ell}{\ell \times Y (\Delta\ell)^2 A} = \frac{2\rho S \ell}{\alpha (\Delta\ell) Y}$$

$$= 500$$

5. **Ans (3)**

For image formed by lens

$$\frac{1}{v_1} - \frac{1}{-15} = \frac{1}{10} \Rightarrow v_1 = +30\text{cm}$$

i.e. 20 cm behind mirror For mirror

$$\frac{1}{v_2} + \frac{1}{20} = \frac{1}{-20} \Rightarrow v_2 = -10\text{ cm}$$

Overall magnification =

$$\left(\frac{30}{-15} \right) \times \left(\frac{10}{20} \right) = 1$$

$$\text{Length of image} = 1 \times 3 = 3\text{ mm}$$

6. **Ans (2)**

$$d\phi = (2\pi x dx) k x t^2$$

$$\phi = \frac{2\pi}{3} k x^3 t^2$$

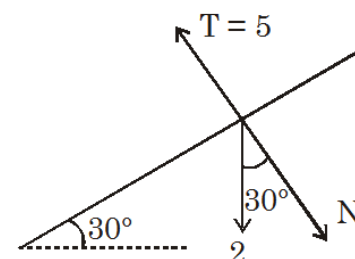
$$E = 2\pi x = \frac{4\pi k x^3 t}{3}$$

$$E = \frac{2k t x^2}{3}$$

$$d\tau = \left(\frac{Q}{\pi R^2} 2\pi x dx \right) \frac{2}{3} k t x^2 x$$

$$\tau = \frac{4k t Q}{3R^2} \int_0^R x^4 dx = \frac{4k t Q R^3}{15}$$

7. **Ans (8)**

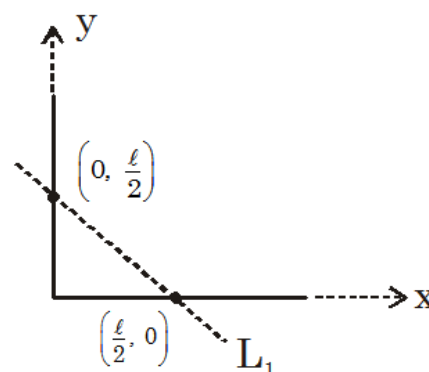


$$N + \sqrt{3} = 5$$

$$f\ell = \mu N$$

$$\Rightarrow 1 = \mu (5 - \sqrt{3}) \Rightarrow \mu = \frac{1}{5 - \sqrt{3}}$$

8. **Ans (4)**



Centre of mass must lie on line L_1

$$\frac{x}{\frac{\ell}{2}} + \frac{y}{\frac{\ell}{2}} = 1$$

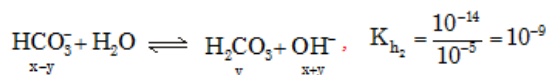
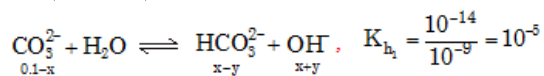
$$2x + 2y = \ell$$

$$a + b = 4$$

CHEMISTRY

SEC-1

1. Ans (A,B,C)



Neglect y w.r.t. x, then

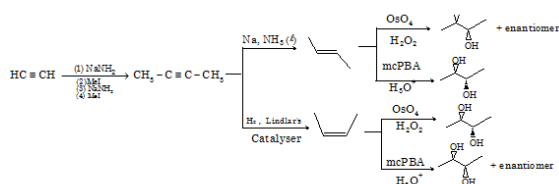
$$K_{h_1} = 10^{-5} = \frac{x^2}{0.1-x} \approx \frac{x^2}{0.1} \Rightarrow x \approx 10^{-3}$$

$$y = K_{h_2} = 10^{-9} = [\text{H}_2\text{CO}_3]$$

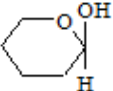
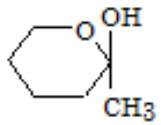
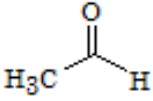
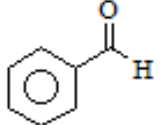
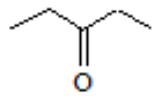
$$[\text{OH}^-] \approx x = 10^{-3} \text{ M} \Rightarrow \text{pOH} = 3, \text{pH} = 11$$

$$[\text{CO}_3^{2-}] \approx 0.1 \text{ M}$$

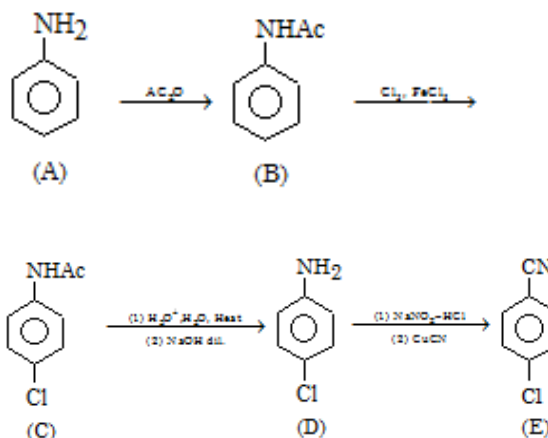
3. Ans (A,B,C)



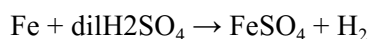
4. Ans (A,B,D)

Comp.	Fehling's Test	Tollen's test
HCHO	Y	Y
	N	N
	Y	Y
	N	N
	Y	Y
	N	Y
	N	N

5. Ans (A,C,D)



8. Ans (A,C)



CHEMISTRY

SEC-2

1. Ans (C)

At 27°C

$$E_{\text{cell}} = 0.4 - 2 \times 10^{-6} (300)^2 = 0.4 - 0.18 = 0.22 \text{ V}$$

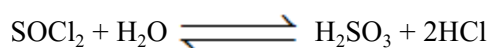
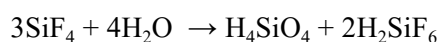
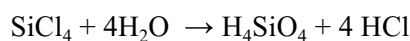
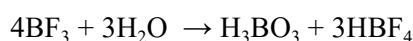
$$\Delta G = -0.22 \text{ F}$$

2. Ans (A)

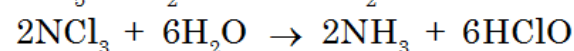
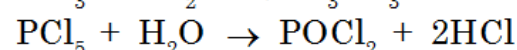
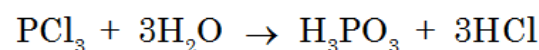
$$\left[\frac{dE_{\text{cell}}}{dT} \right]_P = -4 \times 10^{-6} \times T = -1.2 \times 10^{-3} \text{ V/K}$$

$$\Delta S = 1 \times F \times (-1.2 \times 10^{-3}) = -115.8 \text{ J/K}$$

3. Ans (A,C)



4. Ans (A,B,D)



CHEMISTRY

SEC-3

1. **Ans (9)**

Initial mass ratio of H_2 & $He = 1 : 4$

$\Rightarrow$ Initial molar ratio of H_2 & $He = 1 : 2$

Final molar ratio of H_2 and $He = 8 : 1$

we have,

$$\left(\frac{n_{H_2}}{n_{He}} \right)_{n\text{-step}} = \left(\frac{n_{H_2}}{n_{He}} \right)_{\text{initial}} \left(\sqrt{\frac{M_{He}}{M_{H_2}}} \right)^n$$

$$\frac{8}{1} = \frac{1}{2} \left(\sqrt{\frac{4}{2}} \right)^n \Rightarrow n = 8$$

$\Rightarrow$ minimum number of containers required = $n + 1 = 9$

2. **Ans (3)**

$G = H - TS$

As process is reversible adiabatic the entropy stays constant.

$\Rightarrow dG = dH - SdT$

$$\Delta G = \Delta H - S\Delta T = nC_{p,m}\Delta T - S\Delta T = 1 \times \frac{7}{2} \times R\Delta T - S\Delta T$$

$$= \frac{7}{2} \times 2 \times \Delta T - 3\Delta T = 4 \times \Delta T$$

for adiabatic process

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_2 = T_1 \times \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 1000 \times \left(\frac{1}{32} \right)^{7/5-1} = 250 \text{ K}$$

$$\Delta T = 250 - 1000 = -750$$

$$\Rightarrow \Delta G = 4 \times 4T = 4 \times -750 = -3000 \text{ cal}$$

$$\Rightarrow |\Delta G| = 3 \text{ kcal}$$

3. **Ans (2)**

$$\frac{d}{dT}(\ln K) = \frac{E_a}{RT^2}$$

$$\int d(\ln K) = \int \frac{E_a}{RT^2} \cdot dT$$

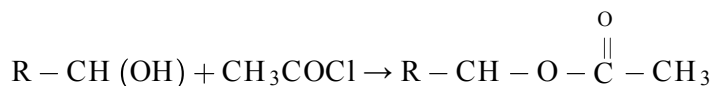
$$\ln K = \int \frac{0.01T^2 + \frac{20}{7}T}{RT^2} \cdot dT$$

$$\ln K = \frac{0.01}{R} \cdot T + \frac{20}{7R} \ln T + C$$

$$\Rightarrow \ln \frac{k_2}{k_1} = \frac{0.01}{R}(400 - 200) + \frac{20}{7R} \ln \frac{400}{200}$$

$$= 1 + 1 = 2$$

4. **Ans (4)**



for every acetylation reaction, molecular mass increases by 42

$$\text{No. of OH group} = \frac{318 - 150}{42} = 4$$

5. **Ans (1)**

Only (f)

6. **Ans (6)**

$$8 - 2 = 6$$

7. **Ans (3)**

$SO_2, SO_3, N(SiH_3)_3$

8. **Ans (6)**

$PbCrO_4, BaCrO_4, (NH_4)_3PO_4 \cdot 12MoO_3,$
 $K_3[Co(NO_2)_6]AgI, PbI_2$

MATHEMATICS

SEC-1

1. Ans (A,C)

$$\begin{aligned} f(x) &= \\ 1 - (\sin 2x) \sqrt[4]{2(\tan^2 x + 3) + \tan^4 x - 5} &= \\ = 1 - (\sin 2x) \sqrt[4]{\tan^4 x + 2\tan^2 x + 1} &= \\ = 1 - (\sin 2x) \sqrt[4]{(\tan^2 x + 1)^2} &= \\ = 1 - (\sin 2x) \sec x &= \\ = 1 - 2 \sin x \in [1, 3] \end{aligned}$$

2. Ans (A,C)

tangents are perpendicular
so they intersect on $M(2, 0)$ lie on directrix
mid point of $A(1, 1)$ and $B(4, 2)$ is $\left(\frac{5}{2}, \frac{3}{2}\right)$
slope of axis is $= \frac{\frac{3}{2} - 0}{\frac{5}{2} - 2} = 3$
equation of directrix, $y - 0 = -\frac{1}{3}(x - 2)$
 $\Rightarrow 3y + x = 2$
 $AM = AS$ (S is the focus and M is the foot of perpendicular from A) $\Rightarrow \frac{2}{\sqrt{10}}$
 $BN = BS$ (S is the focus and N is the foot of perpendicular from B) $\Rightarrow \frac{8}{\sqrt{10}}$
 $AS : BS = 1 : 4$
 $\Rightarrow S\left(\frac{1 \times 4 + 4 \times 1}{5}, \frac{1 \times 2 + 4 \times 1}{5}\right) = \left(\frac{8}{5}, \frac{6}{5}\right)$
solve equation of directrix and eq. of axis to get foot of directrix. vertex is mid point of foot of directrix and focus.

3. Ans (A,C,D)

$${}^{2n}C_n \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n = \sum_{r=0}^n \left(\frac{{}^nC_r}{2^n}\right)^2 = \quad (C)$$

option

$$\begin{aligned} {}^{2n}C_n \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n &= \frac{2n!}{n!n!2^{2n}} \\ &= \frac{2^n \cdot n!1.3.5 \dots (2n-1)}{n!n!2^{2n}} = \frac{1.3.5 \dots (2n-1)}{n!2^n} \end{aligned}$$

$$\prod_{r=1}^n \left(\frac{2r-1}{2r}\right) = (A) \text{ option}$$

$$\prod_{r=1}^n \left(\frac{n+r}{2r}\right) = \frac{(n+1)(n+2)(n+3) \dots (2n)}{2.4.6 \dots 2n}$$

$$= \frac{2n!}{n!2^n \cdot n!} = \frac{{}^{2n}C_n}{2^n}$$

= (B) option wrong

$$\frac{\sum_{r=0}^n ({}^nC_r)^2}{\left(\sum_{r=0}^{2n} {}^{2n}C_r\right)} = \frac{({}^nC_0)^2 + ({}^nC_1)^2 + \dots + ({}^nC_n)^2}{2^n C_0 + \dots + 2^n C_n} = \frac{{}^{2n}C_n}{2^{2n}}$$

= option (D)

4. Ans (A,B,C)

$$P(2^2.5^2) = 2^2.5^2 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 2.5.4$$

$$\therefore \text{no. of numbers multiple of } 2 = 2.5^2$$

$$\text{no. of numbers multiple of } 5 = 2^2.5$$

$$\text{no. of numbers multiple of both} = 2.5$$

$$\therefore P(2^2.5^2) = 2^2.5^2 - 2.5^2 - 2^2.5 + 2.5$$

$$= 2^2.5^2 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right)$$

Similarly check all options.

5. Ans (A,B)

Let the equation of circle 's' is given by

$$(x-1)^2 + (y-1)^2 + \lambda(x+y-2) = 0 \dots (1)$$

sine circle 'x' is orthogonal to the circle

$$x^2 + y^2 + 2x + 2y - 2 = 0 \text{ then } \lambda = 1$$

$$\Rightarrow \text{equation of 's' is } x^2 + y^2 - x - y = 0$$

$\Rightarrow$ now the length of the tangent from $(2, 2)$ is

$$\sqrt{5} = 2.$$

and the equation of the director circle is

$$2x^2 + 2y^2 - 2x - 2y - 1 = 0.$$

6. Ans (B,D)

$$I_m = \int_0^{m\pi} x |\sin x| e^{\cos 4x} dx \dots (1)$$

by using the property

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$I_m = \int_0^{m\pi} (m\pi - x) |\sin x| e^{\cos 4x} dx \dots (2)$$

adding eq. (1) & eq. (2) we get

$$2I_m = m\pi \int_0^{m\pi} |\sin x| e^{\cos 4x} dx$$

by using periodic property

$$2I_m = m^2 \pi \int_0^{\pi} |\sin x| e^{\cos 4x} dx$$

$$2I_m = 2m^2 \pi \int_0^{\pi/2} \sin x e^{\cos 4x} dx$$

$$\Rightarrow I_m = m^2 \pi \int_0^{\pi/2} \sin x e^{\cos 4x} dx \dots (3)$$

and

$$J_n = n \int_0^{\pi} |\cos x| e^{\cos 4x} dx = 2n \int_0^{\pi/2} \cos x e^{\cos 4x} dx \dots (4)$$

on dividing eq. (3) and (4)

$$\frac{I_m}{J_n} = \frac{m^2 \pi}{2n} \text{ now B,D is correct}$$

7. Ans (A,B)

$$\text{Since } f(x) = \cos x \sin 2x = 2 \sin x - 2 \sin^3 x$$

so $\min f(x) = \min g(t)$ where

$$g(t) = 2t - 2t^3 \text{ where } t \in [-1, 1]$$

Now for maxima/minima, we have

$$g'(t) = 2 - 6t^2 = 0 \Rightarrow t = \pm \frac{1}{\sqrt{3}}$$

$$\text{Again } g''(t) = -12t$$

$$\therefore g''\left(\frac{1}{\sqrt{3}}\right) < 0 \text{ \& } g'\left(\frac{-1}{\sqrt{3}}\right) > 0$$

hence

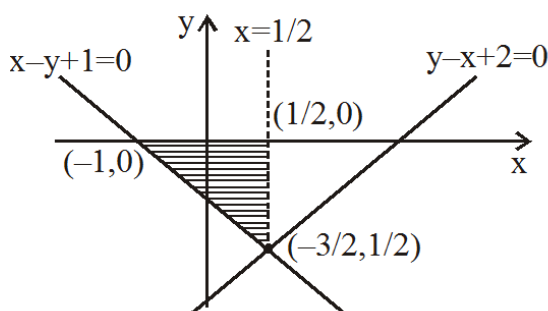
$$\min g(t) = g\left(\frac{-1}{\sqrt{3}}\right) = -\frac{4}{3\sqrt{3}} > \frac{-7}{9} > \frac{-9}{7}$$

8. Ans (B,C,D)

$$\text{Domain } x + y + 1 > 0,$$

$$y - x + 2 > 0$$

$$\text{Squaring } x + y + 1 < y - x + 2$$



$$\text{i.e. } x < \frac{1}{2}$$

Shaded region indicated in the figure

$$\text{and its area} = \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{3}{2} = \frac{9}{8}$$

MATHEMATICS

SEC-2

1. Ans (B)

$$(AB + BA)^T = (AB)^T + (BA)^T$$

$$= B^T A^T + A^T B^T = BA + AB$$

$$(AB - BA)^T = (AB)^T - (BA)^T$$

$$= B^T A^T - A^T B^T = BA - AB = -(AB - BA)$$

2. Ans (C)

$$\text{Since } A \text{ is skew-symmetric matrix therefore } |A| = 0$$

3. Ans (A,B,C)

Using the concept of family of planes let required equation of plane is
 $p_1 + \lambda p_2 = 0$, since the plane passes through
 $(3, 2, 1)$ therefore
 $\Rightarrow 2x - y + z - 2 + \lambda(x + 2y - z - 3)$
 $\Rightarrow \lambda = -1$,
 required plane is $x - 3y + 2z + 1 = 0$

4. Ans (A)

Equation of the plane passes through
 $(-1, 3, 2)$
 is $a(x + 1) + b(y - 3) + c(z - 2) = 0$(1)
 (1) is perpendicular to $L = 0$,
 (Dr's of L are $(2\hat{i} - \hat{j} + \hat{k}) \times (\hat{i} + 2\hat{j} - \hat{k})$) we
 get
 $\frac{a}{-1} = \frac{b}{3} = \frac{c}{5}$ therefore equation of plane is
 $(\vec{r} - (-\hat{i} + 3\hat{j} + 2\hat{k})) \cdot (\hat{i} - 3\hat{j} - 5\hat{k}) = 0$

MATHEMATICS

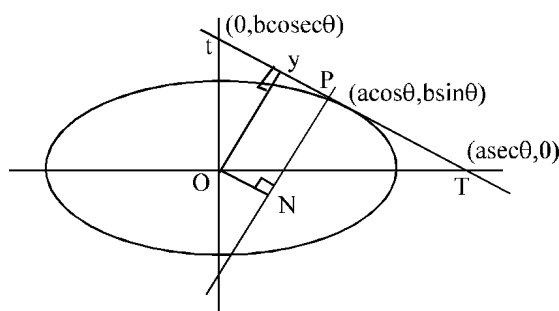
SEC-3

1. Ans (7)

Tangent at P : $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

T : $(a \sec \theta, 0)$, t : $(0, b \operatorname{cosec} \theta)$

hence $Tt = \sqrt{a^2 \sec^2 \theta + b^2 \operatorname{cosec}^2 \theta}$



To find P_y , we draw normal at P.
 now $ON = P_y$

Hence Normal at P

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$$

$$ON : \frac{a^2 - b^2}{\sqrt{a^2 \sec^2 \theta + b^2 \operatorname{cosec}^2 \theta}} = P_y$$

Hence $P_y \cdot Tt = a^2 - b^2$

2. Ans (2)

$$\frac{f(x)}{1+x^2} = 1 + \int_0^x \frac{f^2(t)}{1+t^2} dt \Rightarrow f(0) = 1$$

differentiate w.r.t. 'x' :

$$\frac{(1+x^2) \cdot f'(x) - 2x f(x)}{(1+x^2)^2} = \frac{f^2(x)}{1+x^2}$$

$$\frac{dy}{dx} - \left(\frac{2x}{1+x^2} \right) y = y^2$$

$$\text{or } \frac{1}{y^2} \frac{dy}{dx} - \left(\frac{2x}{1+x^2} \right) \frac{1}{y} = 1$$

$$\text{Let } -\frac{1}{y} = t \Rightarrow \frac{1}{y^2} \frac{dy}{dx} = -\frac{dt}{dx}$$

$$\frac{dt}{dx} + \left(\frac{2x}{1+x^2} \right) t = 1$$

Solution of above equation is

$$-\frac{1}{y} (1+x^2) = \frac{x^3}{3} + x + c.$$

Since $f(0) = 1$, we get $c = -1$

$$\text{Hence, } f(x) = \frac{-3(1+x^2)}{x^3 + 3x - 3}.$$

$$f\left(\frac{1}{2}\right) = \frac{30}{11} \text{ units.}$$

3. Ans (2)

$$P(x) - P(x+1) = 2x - 1 \quad \forall x \in \mathbb{R}$$

$$P(x) = ax^2 + bx + c$$

$$ax^2 + bx + c - [a(x+1)^2 + b(x+1) + c] = 2x - 1$$

$$-[ax^2 + 2ax + a + bx + b]$$

$$-2ax - (a+b) = 2x - 1$$

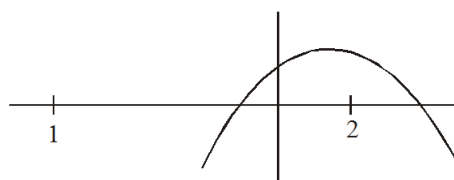
$$-2a = 2 \quad a+b = 1$$

$$a = -1 \quad b = 2$$

$$P(x) = -x^2 + 2x + c$$

Greatest value of

$P(x)$ in $(-\infty, 0]$ is 2



$$\therefore P(0) = c = 2$$

$$\therefore P(x) = -x^2 + 2x + 2$$

$$P(1) = -1 + 2 + 2 = 3$$

$$P(-1) = -1 - 2 + 2 = -1$$

$$P(1) + P(-1) = 2$$

4. Ans (1)

$$S_n = \frac{1}{n} \left(\frac{1}{\sqrt{4n^2}} + \frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-2}} + \dots + \frac{1}{\sqrt{4n^2-(n-1)}} \right)$$

$$L = \lim_{n \rightarrow \infty} (n \cdot S_n)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{4n^2}} + \frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-2}} + \dots + \frac{1}{\sqrt{4n^2-(n-1)}} \right)$$

By sandwich Theorem $L = \frac{1}{2}$

$$M = \lim_{n \rightarrow \infty} (n \cdot S_n)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{4n^2}} + \frac{1}{\sqrt{4n^2-1^2}} + \frac{1}{\sqrt{4n^2-2^2}} + \dots + \frac{1}{\sqrt{4n^2-(n-1)^2}} \right)$$

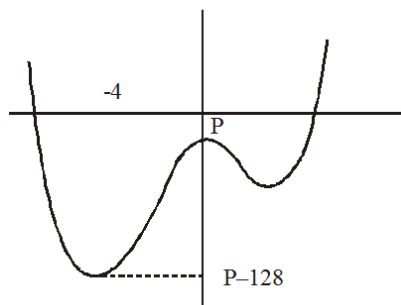
$$= \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{\sqrt{4n^2-r^2}} = \int_0^1 \frac{dx}{\sqrt{4-x^2}} = \left(\sin^{-1} \frac{x}{2} \right)_0^1$$

$$= \frac{\pi}{6}$$

$$\therefore L + M = \frac{1}{2} + \frac{\pi}{6} = \frac{3+\pi}{6} \Rightarrow [L + M] = 1$$

5. Ans (8)

Graph of $f(x)$ is
so for differentiable



$$P - 128 \geq 0$$

$$P \geq 128$$

$$\lambda = 128$$

6. Ans (6)

$$\because (x, 1), (y, 2), (z, 1), (2, 3) \in A \times B$$

$$\Rightarrow A = \{x, y, z, 2\}$$

$$B = \{1, 2, 3\}$$

$$\text{Number of onto map} = {}^4C_2 \cdot 3! = 36$$

7. Ans (4)

$$\left((\vec{\ell} \times (\vec{m} \times \vec{\ell})) \times (\vec{n} \times \vec{m}) \right) \cdot \vec{n}$$

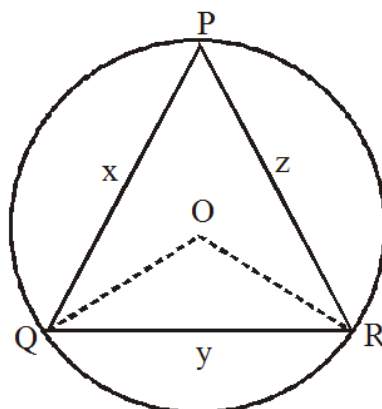
$$\left((\vec{\ell} \cdot \vec{\ell}) \vec{m} \times (\vec{n} \times \vec{m}) \right) \cdot \vec{n}$$

$$|\vec{\ell}|^2 (|\vec{m}|^2 \vec{n}) \cdot \vec{n} = |\vec{\ell}|^2 |\vec{m}|^2 |\vec{n}|^2 = 36$$

$$9 \times \frac{1}{4} \times |\vec{n}|^2 = 36$$

$$|\vec{n}| = 4$$

8. Ans (1)



$$x^2 + y^2 + z^2 = 5$$

$$\angle QPR = 30$$

$$\frac{1}{2} xz \cdot \frac{1}{2} = \frac{\sqrt{3}}{4} \Rightarrow xz = \sqrt{3}$$

$$\cos \angle QPR = \frac{x^2 + z^2 - y^2}{2xz}$$

$$\frac{\sqrt{3}}{2} = \frac{5 - 2y^2}{2 \times \sqrt{3}} \Rightarrow 2y^2 = 2 \Rightarrow y = 1$$

$$\text{from sine rule} = \frac{y}{\sin 30} = 2R \Rightarrow R = 1$$